

Project Proposal: Smartphone Gait-Based Owner Verification

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May 1, 2026

1 Preliminary Domain Knowledge

During our preliminary research, we were especially interested in machine learning projects that make use of data already collected by everyday devices. Smartphones stood out as a strong example because they continuously record information from built-in sensors such as accelerometers and gyroscopes. This suggests that phones may support useful security or health-related applications without requiring extra hardware. For our project, we want to apply this idea to a practical security problem: determining whether a phone remains under its owner’s control while it is being carried by someone who is walking. This matters because theft or unauthorized access does not always happen in an obvious way. A phone might be taken and walked away with, or it might be picked up briefly for a nearby action such as making a payment and then returned. In either case, identifying only the exact moment the phone is taken is usually not enough.

Our main idea is that a person’s gait may act as a behavioral fingerprint. A phone carried while walking records repeated motion patterns, and we believe these signals may contain user-specific information such as stride timing, acceleration structure, and frequency patterns. We found a previous study and repository that approached a related problem as a multiclass classification task, which suggests that different people may produce distinguishable motion signatures with high predictive accuracy. Based on this, our project aims to model walking sequences from smartphone accelerometer and gyroscope data and map them into an embedding space. We then plan to use cosine similarity or another comparison method to estimate whether a new walking segment is likely to come from the phone’s owner or from someone else. In other words, the machine learning task is open-set owner-versus-non-owner verification based on gait-related sensor data collected during walking.

2 Preliminary Data Exploration

The dataset we found on GitHub is the one released with the paper by Zou et al. [3] and its source code repository [4]. It contains 3-axis gyroscope readings (GYR) and 3-axis accelerometer readings (ACC) over time, where Figure 1 shows the sensor axes. Figure 2a shows an example signal. This gives us the channels ACC_x , ACC_y , ACC_z , GYR_x , GYR_y , and GYR_z , each sampled at 50 Hz. As an attempt to filter out the high-frequency noise, we applied a low-pass filter to the raw signals, as shown in Figure 2b. Both raw and smoothed signals show a periodic pattern that is expected from the previous domain knowledge of the walking motion. Hence, we will extract features from frequency domain. Since we frame the task as open-set verification, class balance during training can be controlled by balancing genuine and impostor comparisons within each fold.

Other possible features we consider are the absolute acceleration, which is computed as $\sqrt{ACC_x^2 + ACC_y^2 + ACC_z^2}$, and absolute angular velocity, which is computed as $\sqrt{GYR_x^2 + GYR_y^2 + GYR_z^2}$. These features are expected to be informative because they capture the overall magnitude of acceleration and angular velocity regardless of direction, which can be indicative of the intensity and dynamics of the walking motion.

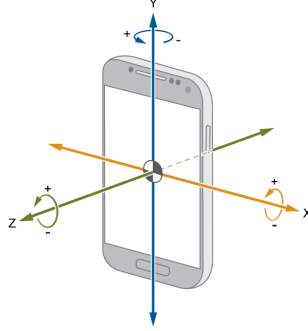


Figure 1: Sensor axes for accelerometer and gyroscope readings [2]

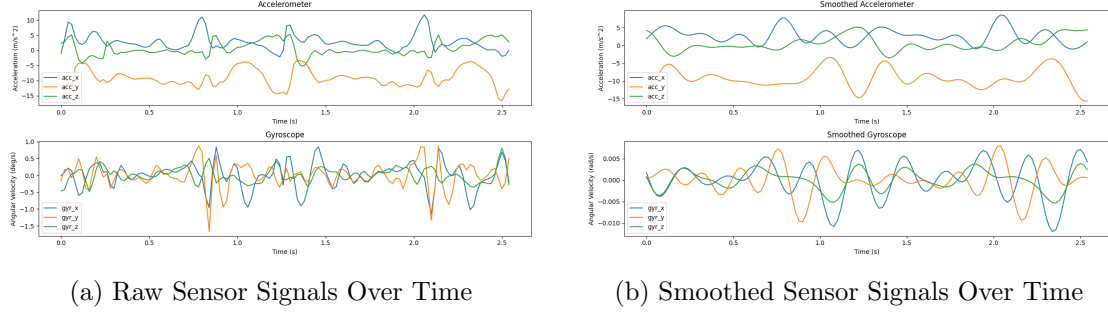


Figure 2: Time-series sensor signal examples

We also computed the correlation matrix of the six sensor channels, which is shown in Figure 3. Even though majority of absolute value of channel correlations are close to zero (≈ 0.1), there can be seen a slight positive correlation of 0.355 between GYR_x and ACC_z . They are predicted to be correlated because both channels are influenced by the inward motion of the leg during stepping forward, where GYR_x captures the femur rotation and ACC_z captures the vertical acceleration from leg lifting.

Accompanying it, there is a negative correlation of -0.229 between GYR_z and ACC_x , which can be explained by the swinging motion of the leg during walking. Lastly, the correlation of 0.153 between ACC_z and ACC_y is believed to be, again, due to the inward motion of the leg during stepping forward. It should be noted that these correlations are weak and could be due to random chance.

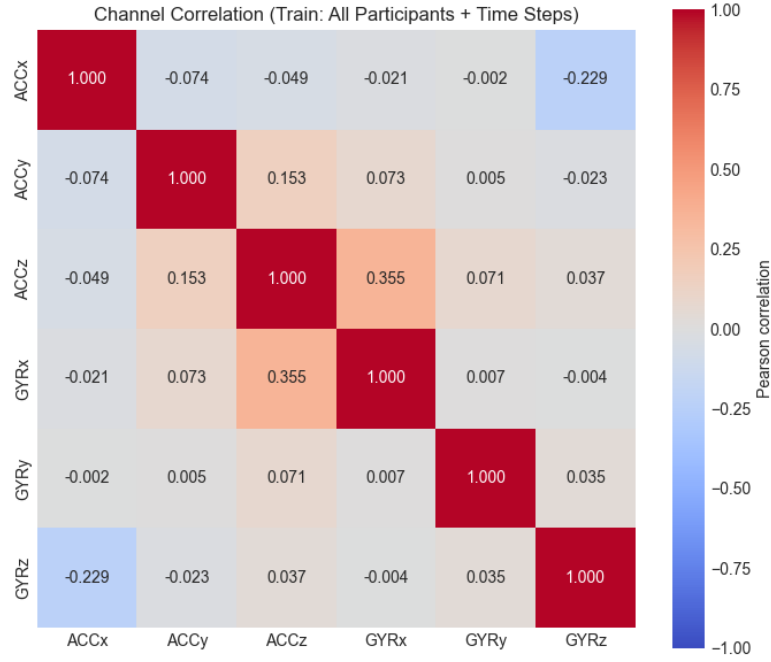


Figure 3: Correlation Matrix of Sensor Readings

As a dimensionality reduction attempt, we applied t-SNE, with 118 classes (one for each embedding in our dataset), to the six sensor channels, which is shown in Figure 4. First thing to notice here is that the peripheral clusters are well separated from each other, indicating easier classification of those samples. However, the central cluster is mixed and dense, which is expected for a non-linear model to separate these classes.

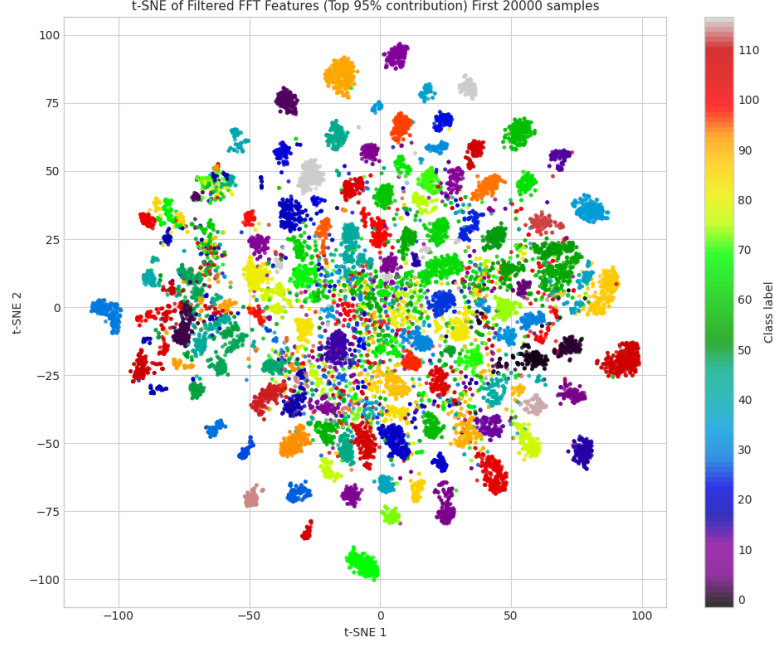


Figure 4: t-SNE plot of the dataset

There is no missing data in the dataset. Regarding outliers, we observe that the sensor readings fall outside typical ranges, particularly in the gyroscope channels. However, we note that outliers in gait data may represent distinctive behavioral patterns that are part of an individual's biometric fingerprint rather than measurement errors. As a behavioral biometric, gait naturally exhibits individual variation, including asymmetries and characteristic motion anomalies. We will explore the impact of outlier removal during model development but expect to retain most outliers in our final feature set, as they may be diagnostic of person-specific gait characteristics.

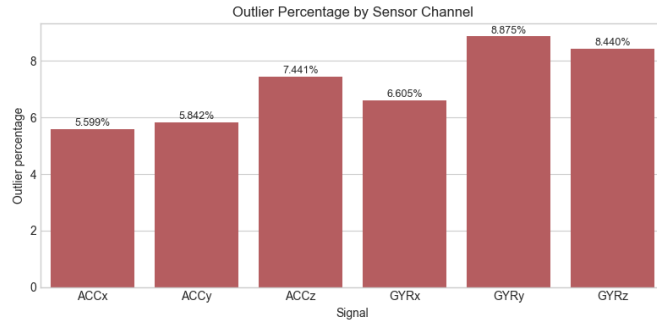


Figure 5: Percentage of outliers in the different channels, calculated as the greater than 1.5 IQR from the mean of each channel.

3 Proposed Preprocessing

We would like to propose a preprocessing pipeline that is organized into two stages. First, we will apply a shared pipeline for all experiments, starting with data quality checks. Since we have confirmed that there are no missing values, we will examine outliers and sensor artifacts (flatline sections or implausible accelerometer/gyroscope spikes). However, rather than automatically removing all detected outliers, we will take a principled approach: we will test the impact of outlier removal on model performance, but we expect to retain most outliers in our feature set. This is motivated by behavioral biometrics literature on gait recognition, which emphasizes that outliers may encode distinctive person-specific patterns. As noted in the gait biometrics survey, it is crucial to recognize that “on gait data...it underlines how crucial it is to group gait data according to a person’s health in order to identify irregularities and classify them” [1]. In the context of open-set verification, extreme values in acceleration or rotation may represent individually distinctive motion characteristics that help distinguish the owner from impostors. Segmentation has already been performed on the raw data, and each sample will be represented as a segmented window. The original paper’s segmentation strategy uses the aforementioned cumulative acceleration to identify each two-step window, which are of variable length. To create a fixed-length representation, they apply linear interpolation to produce windows of length 128. Many papers, including the original, have found that meaningful segmentations improve predictive accuracy, so we will use this baseline data.

In a different approach than the original paper, we will use strict subject-level splitting to avoid identity leakage. We think this is the safest approach, with clean separation maintained throughout by participant. We will use a 90/10 data split for train/test, with the training set used to construct $k \sim 5$ folds by subject groups with 80/20 for training and validation, which gives us good averaging across validation folds. The full training set will be used later. This depends on the training intensity we find; we may adjust the k value accordingly. We will use per-channel z-score normalization computed only on the training partition and then reused for validation and test data. We will have to precompute the folds since the z-score will be specific for the validation run, then the test set z-score computed on the full training set statistics.

After the shared stage, we branch by model type. For the baseline model, we will prepare manually computed frequency-domain features per window (FFT cumulative energy contribution), and then perform model-specific feature selection during baseline testing. We have already experimented with FFT features ranked by importance to prediction with success. For the main deep-learning track, we will test both whether feature selection benefits the deeper model and raw data. As an important distinction, we have chosen not to construct triplet pairs during preprocessing. This is motivated by wanting to test different sampling strategies, such as random selection, hard negative mining, and batch-level negatives, and to decide later based on efficiency.

4 Proposed Model(s) and Baseline

For the baseline, we will use a nearest-centroid or prototype-based classifier on the FFT features from preprocessing. This gives us a simple and cheap reference point that is easy to interpret and useful as a comparison against the transformer model. For the main model, we plan to use a transformer because it can learn non-linear relationships across the time dimension of the gait windows instead of relying only on hand-crafted summary features. The transformer will be trained with triplet loss as the primary training objective, so that windows from the same person move closer together in embedding space and windows from different people move farther apart. In addition, we will explore other metric-learning optimization techniques as extensions, while keeping triplet loss as the main loss function.

We will use dropout, L2 regularization, and normalization layers to reduce overfitting. Early stopping will be based on the validation equal error rate (EER), since that is the clearest summary of verification performance for this task. These choices should also make the model easier to compress later, because a less overfit model is usually less sensitive when weights are pruned or quantized. For deployment, we want two levels of success, one for the base grade and one extra. The baseline goal is to expose the model through an API endpoint so the project is functional and can return predictions from sensor input. The extra goal is on-device deployment, so the phone itself can run the model locally with as little dependency on a server as possible. This would be the more applied version of the project and is the main stretch goal.

5 Proposed Evaluation

For the main open-set verification task, we will evaluate the model primarily with false acceptance rate (FAR), false rejection rate (FRR), and equal error rate (EER), because these metrics directly reflect how well the system accepts genuine users and rejects impostors. We may still report accuracy, precision, recall, F1 score, confusion matrices, and AUROC as secondary metrics when they help with comparison, but these are less central to the verification setting than FAR and FRR.

We also want to understand what the model is learning, not only how well it performs. For the FFT-based baseline, we can inspect which frequency components are most useful for separating different pocket placements and visualize them with bar plots or heatmaps. For the transformer model, we want to visualize the embedding space to see how well the model separates users, and, if possible, we may also use additional interpretability tools to inspect what parts of the input are driving its decisions.

Finally, we will evaluate the model on new data that we record from a few volunteers who were not part of the original dataset. They will carry the phone in different pockets while walking, and we will use this data to check how well the trained model generalizes to new people and to more realistic deployment conditions. This will give us a more practical sense of whether the system is robust outside the original training data. Given our deployment plan, this will initially mean sending data to our endpoint, but eventually it will mean live use on a simple app on a phone.

6 Model Usage

As hypothesised in the introduction and based on preliminary domain knowledge, the model could be used for theft and suspicious activity detection, purchase confirmation based on recent motion behaviour, and other lightweight security checks requiring identity confirmation. While these are all practical applications, they are feasible and realistic due to the prospective system being relatively small, capable of running on a device and having access to live motion data. The input of acceleration and rotation data and the output of an onset/offset verification have a wide range of applications, and the model itself will likely be part of a larger onboard security system. The great thing is that anyone with a smartphone, especially those who want an additional passive layer of security, can use the model. Since our initial goal is to create a working service with just an API endpoint, this can be used for less live and onboard security, similar to software such as FindMy and Life360, which perform passive and possibly interval checks. We don't want to livestream biometric data, so ideally we want edge deployment.

In order for the model to function, there would first be a short enrolment period during which the user would walk for a few minutes in order to create an identification embedding or other enrolment, such as a centroid of features. After that, live data would be streamed in windows based on step timing, as described in the original paper. This data would be interpolated and resampled to handle jitter, and then compared against the known owner embedding using cosine similarity or another comparison method. Most of the time, the system would remain idle after the model. If it detects a potential threat, the app could trigger an alert, message or notification on another device. The final output is essentially a yes/no decision, so the user-facing result would probably be a notification on another device. In order to support this, we would require SMS, email, bot-based messaging or a server that can receive the signal and send a ping to another device. This could act as theft detection for children with phones, with parents receiving notifications when their child gives the device to an untrusted adult or it is stolen.

7 Risk Assessment

Some potential risks related to project development include data leakage across training and test sets, over-complexity of the transformer models for the task, a trade-off between model size and accuracy, and the need to learn new techniques for model compression and deployment.

During development, we may encounter issues on several fronts. As transformer models are often large and computationally expensive, we may end up with an overly-complicated model that is lenient to overfitting. We may also face challenges in handling live streaming sensor data, which is both technically demanding and sensitive in nature. Additionally, deployment constraints may require us to learn techniques specific to this use case, such as mobile development.

To mitigate these risks, we adopt several strategies. To prevent data leakage, as mentioned in previous sections, we will split the data by subject so that no sensor data from a test subject appears in training. To address model complexity, we will start with a smaller transformer architecture and only increase complexity if necessary, using the baseline model as a reference. We will also apply regularization techniques such as dropout and early stopping to prevent overfitting.

To prepare for deployment challenges early, we plan to study projects such as the web **fake beer** drinking app to understand how live streaming sensor data is handled. Finally, to minimize privacy risks from live data, we would prefer to keep the system on-device rather than stream sensor data to a server, thereby reducing the risk of data breaches and privacy violations.

The project is likely to be successful in terms of model accuracy, since the task has already been shown to be feasible with high accuracy. What is less certain is how viable the approach will be once compression and deployment constraints are introduced.

The main societal risk is misuse: The system could be used to detect, follow, or identify someone through their embedding, which would effectively leak biometric information. There are also user risks because this system would serve as an added security layer, and users might blame the model if it fails in a sensitive situation. In a real deployment, the model would need to achieve a very high level of reliability before it could be trusted.

8 Individual Learning Outcomes

Will: I would like to learn more about model compression as a general practice and about the nuance of deploying such a model. This connects to my interest in neuromorphic computing, because smaller neural networks are easier to simulate in that setting and can provide useful insight into how these ideas might be implemented in circuitry.

Haukur: I would like to learn to write a high quality report for an applied project, work in a larger team of four for a continued longer project. I believe this entails keeping good organization on github, creating issues and managing the project at a team level. I also find end-to-end model deployment great to understand, as it aligns with my personal and professional goals.

Tijje:

Efe: I would like to learn how to deploy a machine learning model in a real-world setting. Plus, learning to work with continuous sensor data overlaps with my interest in robotics. Picking up teamwork and github management skills in larger teams is also a key learning outcome for industry readiness. As I would like to convert what I learn in this course into industrial value.

9 Relation to Grading Specifications

Our preprocessing plan is to start with the segmented windows, check for sensor artifacts and outliers, normalize each channel using training-data statistics, and then prepare either raw windows or engineered features depending on the model. For cross-validation, we will use subject-level splits so that data from the same person does not appear in both training and testing, which better measures generalization to new users. For optimization, we will tune the main hyperparameters of each model, such as learning rate, dropout, regularization strength, and model-specific architecture settings, using validation performance. To evaluate overfitting, we will compare training and validation performance and use regularization methods such as dropout, L2 regularization, and early stopping when needed.

We also see our novelty coming from the in the real-world setting of gait-based owner verification on phone sensor data, especially because we want the model to generalize to unseen people rather than only recognize users it has already seen. In addition, we aim to keep the model lightweight enough for possible smartphone deployment. The project should also satisfies the reproducibility requirements. We have maintained a structured GitHub repository from the start, used feature branches, written clear commit messages, and kept notes and code organized so the workflow can be followed and reproduced.

10 Timeline

The first week corresponds to the week of 20-26 April.

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Project Search/Selection								
Literature Search								
Proposal Writing								
Data Exploration								
Preprocessing								
Training baseline model								
Training main model								
Evaluation								
Model Deployment (API)								
Implementing Extras (Model Compression, Explainability & on-device deployment)								
Report Writing								

Table 1: Proposed timeline for the project.

References

- [1] Vyom Kulshreshtha and KM Nivedita Singh. A survey of behavioral biometric gait recognition: Current success and future directions. *International Journal*, 2024. Essential feature selection for a bipedal robot’s stability and effective operation.
- [2] MathWorks. Gyroscope block [digital image]. MathWorks Help Center, 2014. In *Gyroscope – Measure rotational speed around X, Y, and Z axes in rad/s*. Retrieved April 30, 2026, from https://nl.mathworks.com/help/simulink/supportpkg/android_ref/gyroscope.html.
- [3] Qin Zou, Yanling Wang, Qian Wang, Yi Zhao, and Qingquan Li. Deep learning-based gait recognition using smartphones in the wild. *IEEE Transactions on Information Forensics and Security*, 15:3197–3212, 2020.
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